MIL-C-18312/7B
12 January 1970
SUPERSEDING
MIL-C-18312/7A
7 June 1966

MILITARY SPECIFICATION SHEET

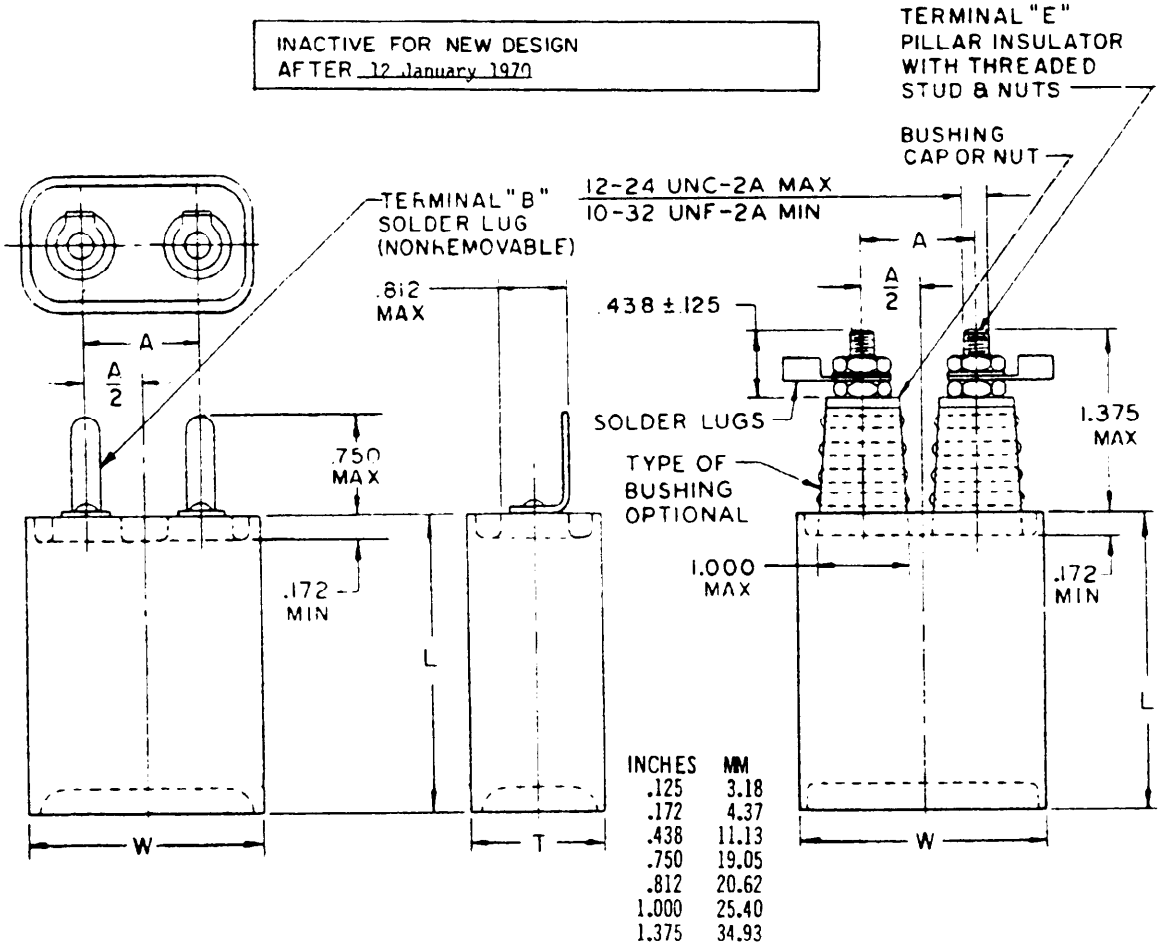
CAPACITORS, FIXED, METALLIZED (PAPER, PAPER PLASTIC, OR PLASTIC FILM) DIELECTRIC, DIRECT CURRENT, (HERMETICALLY SEALED IN METAL CASES)

STYLE CH72

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-18312.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

INACTIVE FOR NEW DESIGN
AFTER 12 January 1970



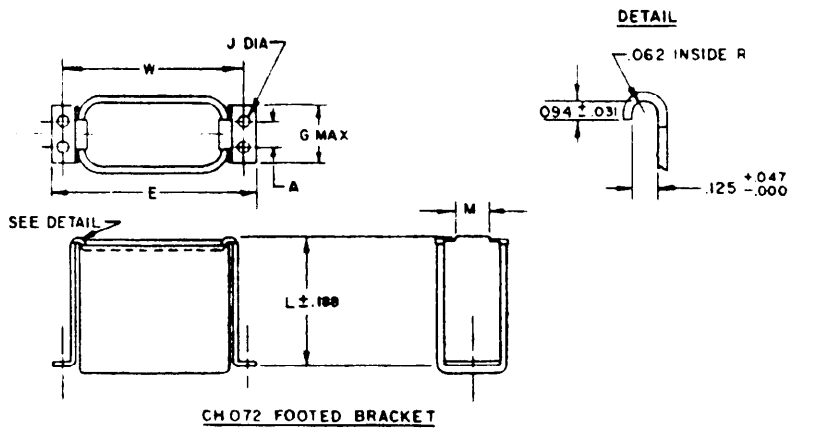
INCHES	MM
.125	3.18
.172	4.37
.438	11.13
.750	19.05
.812	20.62
1.000	25.40
1.375	34.93

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. See table I for additional dimensions.

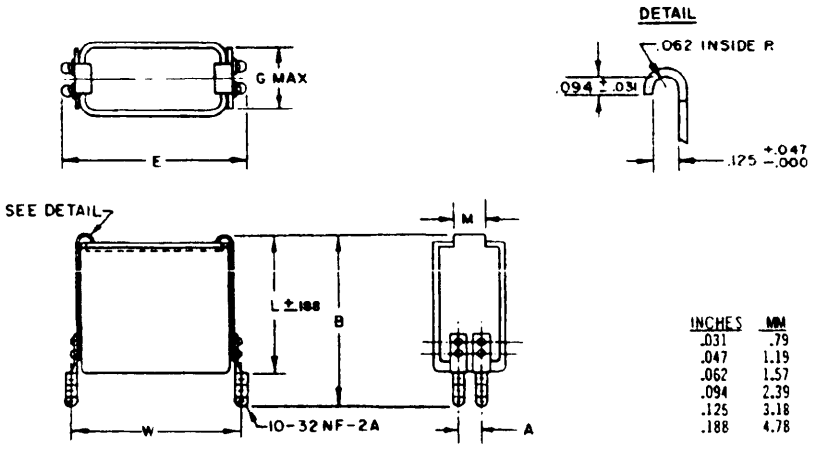
FIGURE 1. Style CH72 capacitor.

FSC 5910

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CH 072 FOOTED BRACKET



INCHES	MM
.031	.79
.047	1.19
.062	1.57
.094	2.39
.125	3.18
.188	4.78

CH 072 SPADE-LUG BRACKET
(FOR USE WITH STYLE CH72 CAPACITORS
CASE SIZES A-1 THROUGH C3)

NOTES:
1. All dimensions in inches.
2. Shape of bracket optional provided specified dimensions are met. Spade lugs shall be rigid and double welded, double riveted, single riveted and soldered, or welded and single riveted to the bracket.
3. Minimum thickness of brackets shall be .045.
4. Brackets shall be marked on the outer surface.
5. Metric equivalents (to the nearest .01 mm) are given for general information only, and are based upon 1 inch = 25.4 mm

FIGURE 2. CH072 brackets.

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REQUIREMENTS:

Dimensions: See figures 1 and 2, and tables I and III.

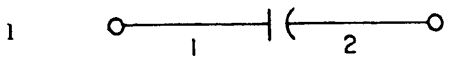
Case material: Magnetic or nonmagnetic.

Capacitance value: See table II.

Capacitance tolerance: See table II.

DC rated voltage: See table II.

Circuit diagram:



Seal:
Method 112 of MIL-STD-202, condition A.
Liquid-filled capacitors - One hour at 125° ±3° C with no evidence of leakage.

Dielectric withstanding voltage:
Method 301 of MIL-STD-202.
Terminal to terminal - 200 percent of rated dc voltage.
Terminal to case 400 percent of rated dc voltage or 2,000 volts, whichever is lower.

Capacitance:
Method 305 of MIL-STD-202 (for value and tolerance see table II).

Dissipation factor:
At +25° C - 1.0 percent.
At +125° C - 2.0 percent.

Insulation resistance:
Terminal to terminal - See table V.
Terminal to case 10,000 megohms.

Barometric pressure:
Method 105 of MIL-STD-202.
Test condition - B.

Vibration, low frequency:
Method 201 of MIL-STD-202.

Temperature cycling:
Method 102 of MIL-STD-202, condition D, except that for step 3, temperature shall be 125° C.

Immersion:
Method 104 of MIL-STD-202, condition C.
Insulation resistance:
Terminal to terminal - Not less than 1/3 the value specified in table V.
Terminal to case - 5,000 megohms, minimum.

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Dielectric withstanding voltage:
Terminal to terminal - 150 percent rated dc voltage.
Terminal to case - 400 percent of rated dc voltage or 2,000 volts whichever is lower.
Capacitance change - Not to exceed 10 percent of initial value.
Dissipation factor - Not to exceed 110 percent of initial requirement.

Terminal strength:
Method 211 of MIL-STD-202, conditions A and E.
Terminal B:
Condition A - Applied force 2 pounds.
Condition E - Torque 2 pound-inches.
Terminal E:
Condition A - Applied force 10 pounds.
Condition E - Torque 15 pound-inches.

Low temperature and capacitance change with temperature.
Conditions - 48 hours with rated voltage applied.
Capacitance change - See table V.

Life:
Qualification - Test voltage 140 percent of rated voltage, 1,000 hours at high ambient temperature.
Quality conformance - 250 hours at conditions specified for 1,000-hour test.
Dissipation factor:
At high ambient test temperature (any time 24 to 48 hours from start of test) - Not greater than 2.0 percent.
At high ambient test temperature (any time during last 48 hours of test). Not greater than 2.5 percent.
At 25° C (after life test) - Not greater than 1.25 percent.

Capacitance:
At 25° C (after life test) - Within 10 percent of initial value.

Insulation resistance: At 25° C (after life test)
Terminal to terminal - Not less than 1/3 value specified in table V.
Terminal to case - Not less than 5,000 megohms.

Supersession information: See tables II and III.

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Purchaser to develop plans for the logistic support analysis (LSA)

TABLE I. *CH72 Case sizes, case dimensions, and bracket designations.*

Case size	Case dimensions 2/ W ±.062	T ±.062	L +.062 -.125	A +.062 - .187	Bracket 1/ designation
A1	1.812	1.062	2.375	0.813	CH072 - A1
A2	1.812	1.062	2.750	0.813	CH072 - A2
A3	1.812	1.062	3.500	0.813	CH072 - A3
A4	1.812	1.062	3.750	0.813	CH072 - A4
A5	1.812	1.062	4.000	0.813	CH072 - A5
A6	1.812	1.062	1.875	0.813	CH072 - A6
B1	2.500	1.188	3.250	1.125	CH072 - B1
B2	2.500	1.188	3.875	1.125	CH072 - B2
B3	2.500	1.188	4.000	1.125	CH072 - B3
B4	2.500	1.188	5.250	1.125	CH072 - B4
C1	3.812	1.250	3.625	2.000	CH072 - C1
C3	3.812	1.250	4.250	2.000	CH072 - C3
C4	3.812	1.750	4.000	2.000	CH072 - C4

1/ See table III for applicable bracket information.

2/ See table IV for metric equivalents.

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TABLE II. *Style CH72 characteristics, case sizes, and superseded dash numbers.*

Type designation 1/	DC rated voltage	Capacitance value	Terminal	Charac-teristic	Case size	Superseded MIL-C-18312/7A dash numbers	
						Capacitance tolerance	
						(K)±10%	(M)±20%
	Volts	*µf*					
CH72B1NV156-	150	15	B	N	A1	0053	0054
CH72E1NV156-	150	15	E	N	A1	0055	0056
CH72B1NV226-	150	22	B	N	A3	0057	0058
CH72E1NV226-	150	22	E	N	A3	0059	0060
CH72B1NV336-	150	33	B	N	B1	0061	0062
CH72E1NV336-	150	33	E	N	B1	0063	0064
CH72B1NV476-	150	47	B	N	B3	0065	0066
CH72E1NV476-	150	47	E	N	B3	0067	0068
CH72B1NE475-	400	4.7	B	N	A3	0069	0070
CH72E1NE475-	400	4.7	E	N	A3	0071	0072
CH72B1NE685-	400	6.8	B	N	A5	0073	0074
CH72E1NE685-	400	6.8	E	N	A5	0075	0076
CH72B1NE106-	400	10	B	N	B2	0077	0078
CH72E1NE106-	400	10	E	N	B2	0079	0080
CH72B1NE156-	400	15	B	N	C1	0081	0082
CH72E1NE156-	400	15	E	N	C1	0083	0084
CH72B1NE226-	400	22	B	N	C3	0085	0086
CH72E1NE226-	400	22	E	N	C3	0087	0088
CH72B1NF475-	600	4.7	B	N	A4	0089	0090
CH72E1NF475-	600	4.7	E	N	A4	0091	0092
CH72B1NF685-	600	6.8	B	N	B1	0093	0094
CH72E1NF685-	600	6.8	E	N	B1	0095	0096
CH72B1NF106-	600	10	B	N	B3	0097	0098
CH72E1NF106-	600	10	E	N	B3	0099	0100
CH72B1NF156-	600	15	B	N	C1	0101	0102
CH72E1NF156-	600	15		N	C1	0103	0104
CH72B1NF226-	600	22	B	N	C4	0105	0106
CH72E1NF226-	600	22	E	N	C4	0107	0108

1/ Complete type designation will include an additional symbol for capacitance tolerance.

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TABLE III. *CH072 bracket dimensions and superseded dash numbers.*

Bracket designation 1/	W ±.032 3/		E max.	G max.	A 2/ ±.016	J	L		B ±.062	M ±.016	Case size	Superseded MIL-C-18312/7A dash number	
	Footed bracket	Spade lug bracket					Footed bracket	Spade lug bracket				Footed bracket	Spade lug bracket
CH072-A1	2.250	2.062	2.718	0.625	NOT APPLICABLE (SINGLE HOLE OR STUD)	0.187	2.250	2.125	3.125	0.391	A1	0801	0802
CH072-A2	2.250	2.062	2.718	0.625		0.187	2.625	2.500	3.500	0.391	A2	0803	0804
CH072-A3	2.250	2.062	2.718	0.625		0.187	3.375	3.250	4.250	0.391	A3	0805	0806
CH072-A4	2.250	2.062	2.718	0.625		0.187	3.625	3.500	4.500	0.391	A4	0807	0808
CH072-A5	2.250	2.062	2.718	0.625		0.187	3.875	3.750	4.750	0.391	A5	0809	0810
CH072-A6	2.250	2.062	2.718	0.625		0.187	1.750	1.625	2.625	0.391	A6	0811	0812
CH072-B1	3.000	2.750	3.938	0.625		0.187	3.125	3.000	4.000	0.391	B1	0821	0822
CH072-B2	3.000	2.750	3.938	0.625		0.187	3.750	3.625	4.625	0.391	B2	0823	0824
CH072-B3	3.000	2.750	3.938	0.625		0.187	3.875	3.750	4.750	0.391	B3	0825	0826
CH072-B4	3.000	2.750	3.938	0.625		0.187	5.125	5.000	6.000	0.391	B4	0827	0828
CH072-C1	4.438	4.062	5.032	0.625		0.187	3.500	3.375	4.375	0.391	C1	0841	0842
CH072-C3	4.438	4.062	5.032	0.625		0.187	4.125	4.000	5.000	0.391	C3	0845	0846
CH072-C4	4.438	4.062	5.032	1.375	0.625	0.213	3.875	3.750	4.750	0.843	C4	0847	0848

1/ Complete bracket designation will include the symbol "F" for footed or "S" for spade lug to indicate design of bracket.

2/ Single hole or spade bolt located within .016 inches of center line of bracket.

3/ See table IV for metric equivalents.

TABLE IV. *Millimeter equivalents of decimal inches.*

Inches	MM	Inches	MM	Inches	MM
.016	.41	1.875	47.63	3.812	96.82
.032	.81	2.000	50.80	3.875	98.43
.062	1.57	2.062	52.37	3.938	100.03
.125	3.18	2.125	53.98	4.000	101.60
.187	4.75	2.250	57.15	4.062	103.17
.213	5.41	2.375	60.33	4.125	104.78
.391	9.93	2.500	63.50	4.250	107.95
.625	15.88	2.625	66.68	4.375	111.13
.813	20.65	2.718	69.04	4.438	112.73
.843	21.41	2.750	69.85	4.500	114.30
1.062	26.97	3.000	76.20	4.625	117.48
1.125	28.58	3.125	79.38	4.750	120.65
1.188	30.18	3.250	82.55	5.000	127.00
1.250	31.75	3.375	85.73	5.032	127.81
1.375	34.93	3.500	88.90	5.125	130.18
1.625	41.28	3.625	92.08	5.250	133.35
1.750	44.45	3.750	95.25	6.000	152.40
1.812	46.02				

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TABLE V. *Characteristics.*

Operating temperature range	-55 to +125° ±3° C	
DC voltage rating (volts)	150	400 600
High ambient test temperature	125° ±3° C	125° ±3° C
Low ambient test temperature	-55° ±3° C	-55° ±3° C
Megohms x microfarads:		
At 25° C	2,000 min	2,000 min
At high ambient test temperature	10 min	40 min
Insulation resistance (megohms):		
At 25° C 1/	12,000	12,000
At high ambient test temperature 1/	150	600
Capacitance change at low temperature	-15% max	-15% max
Life test voltage (% rated voltage)	140%	140%

1/ Need not exceed.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - EC
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - None
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Navy - EC

(Project 5910-0968-8)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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